DNA Structure: The Blueprint of Life

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?? Notes

I. Introduction to DNA

- **Deoxyribonucleic acid (DNA)**: The molecule that carries genetic instructions for all known organisms and many viruses.
- Contains the information for building and maintaining an organism.
- Located in the nucleus of eukaryotic cells and cytoplasm of prokaryotic cells.

II. DNA Components: Building Blocks of the Double Helix

- **Nucleotides**: The monomers (building blocks) of DNA.
- Composed of three parts:
- **Deoxyribose sugar**: A five-carbon sugar.
- **Phosphate group**: Attached to the 5' carbon of the sugar.
- **Nitrogenous base**: Attached to the 1' carbon of the sugar. There are four types:
- **Adenine (A)**: A purine (double-ring structure).
- **Guanine (G)**: A purine.
- **Cytosine (C)**: A pyrimidine (single-ring structure).
- **Thymine (T)**: A pyrimidine.

III. DNA Structure: The Double Helix

- **Double helix**: The three-dimensional structure of DNA, resembling a twisted ladder.
- **Strands**: DNA consists of two strands running antiparallel to each other.
- **Antiparallel**: One strand runs 5' to 3', while the other runs 3' to 5'.
- The 5' end has a phosphate group attached to the 5' carbon of deoxyribose.
- The 3' end has a hydroxyl group (-OH) attached to the 3' carbon of deoxyribose.

- **Sugar-phosphate backbone**: Forms the structural framework of the DNA molecule.
- Formed by phosphodiester bonds between the sugar of one nucleotide and the phosphate group of the next.

- **Base pairing**: The specific hydrogen bonding between nitrogenous bases holds the two strands together.
- **Complementary base pairing**: Adenine (A) always pairs with Thymine (T), and Guanine (G) always pairs with Cytosine (C).
- A-T pairing involves two hydrogen bonds.
- G-C pairing involves three hydrogen bonds (stronger bond).

- **Major and Minor Grooves**: The double helix has grooves that are important for protein binding.
- **Major groove**: Wider and more accessible for protein interaction.
- **Minor groove**: Narrower and less accessible.

IV. DNA Organization in Cells

- **Prokaryotic DNA**: Typically a single, circular chromosome located in the cytoplasm.
- **Eukaryotic DNA**: Organized into multiple linear chromosomes located in the nucleus.
- **Chromatin**: DNA is associated with proteins (histones) to form chromatin.
- **Histones**: Proteins around which DNA is wrapped.
- **Nucleosome**: The basic unit of chromatin, consisting of DNA wrapped around eight histone proteins.
- **Chromosomes**: Tightly packed chromatin during cell division.
- **Sister chromatids**: Two identical copies of a chromosome formed during DNA replication, attached at the centromere.

V. DNA Replication: Copying the Code

- **Semi-conservative replication**: Each new DNA molecule consists of one original strand and one newly synthesized strand.
- **Key enzymes involved**:
- **DNA polymerase**: Adds nucleotides to the 3' end of the new strand, using the existing strand as a template.
- **Helicase**: Unwinds the DNA double helix.
- **Ligase**: Joins DNA fragments together.
- **Primase**: Synthesizes RNA primers to initiate replication.

VI. Significance of DNA Structure

- **Information storage**: The sequence of nucleotides stores genetic information.
- **Replication**: The structure allows for accurate replication of genetic information.
- **Mutation**: Changes in the DNA sequence can lead to mutations, which can have various effects.
- **Gene expression**: The DNA sequence determines the sequence of amino acids in proteins, influencing gene expression.

#?? Key Terms & Definitions

Deoxyribonucleic Acid (DNA): The molecule that carries genetic instructions for all known organisms.

Nucleotide: The monomer of DNA, consisting of a deoxyribose sugar, a phosphate group, and a nitrogenous base.

Nitrogenous Base: A molecule containing nitrogen and having chemical properties of a base (A, T, G, C).

Double Helix: The three-dimensional structure of DNA, resembling a twisted ladder.

Complementary Base Pairing: The specific hydrogen bonding between nitrogenous bases (A with T, and G with C).

Antiparallel: The arrangement of DNA strands where one runs 5' to 3' and the other runs 3' to 5'.

Chromosome: A thread-like structure of nucleic acids and protein found in the nucleus of most living cells, carrying genetic information in the form of genes.

?? Examples

- **Sickle cell anemia**: A genetic disorder caused by a single nucleotide mutation in the gene that codes for hemoglobin.
- **DNA fingerprinting**: Uses the unique DNA sequences of individuals to identify them (e.g., in forensic science).
- **Viruses**: Some viruses use DNA as their genetic material, highlighting the importance of DNA structure in diverse biological systems.

? Review Questions

- What are the three components of a nucleotide?
- Explain the concept of complementary base pairing in DNA.
- Describe the structure of the DNA double helix.
- What is the role of histones in eukaryotic DNA organization?
- Why is DNA replication described as semi-conservative?
- How does the antiparallel arrangement of DNA strands affect replication?
- Explain how mutations in DNA can affect an organism.

?? Summary

DNA is the molecule that carries genetic information, composed of nucleotides arranged in a double helix structure. The double helix consists of two antiparallel strands held together by complementary base pairing (A-T and G-C). Eukaryotic DNA is organized into chromosomes, which are composed of DNA and proteins (histones). DNA replicates semi-conservatively, ensuring accurate transmission of genetic information from one generation to the next. The structure of DNA allows for information storage, replication, and gene expression.